

Christopher R. Culpepper

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EXPERIENCE

Electrical Engineer

October 2019 - Present

FTL Labs Corporation, Amherst MA

- Designed, assembled, tested FPGA RFSoc based RADAR system
 - Included Xilinx RFSoc module, X-Band RF LO, downconverter
 - Fully owned electrical hardware, as well as FPGA code
 - Project included a large amount of reverse engineering and RF design
 - Mechanically designed add-ons for commercial RADAR
 - Iterated design to meet signal integrity, EMI, shock, vibration testing
- Reverse engineered commercial RADAR to control magnetron (UART, GPIO, RF Mixer, LNA)
- Designed and built large li-ion battery for underwater applications, designed with safety in mind
- Designed, implemented acoustic threshold measurement system, with absolute SPL, HW + SW
- Designed LED illumination array for blood clot detection
- Designed interface electronics, analog piezo load cell interface, COTS LED module dimmer
- Personally ran small-scale assembly and production (12 units)
- Reverse engineered commercial laser diode driver to implement circuit

Senior Electrical Engineer

Coop 2013-2017, FT 2017-2019

General Dynamics Mission Systems, Pittsfield MA

- Performed software and systems testing on high-integrity mission-critical software
- Managed requirements and test cases
- Performed integration and repair operations on legacy hardware
- Go-to person in the lab to diagnose and troubleshoot weird hardware and installation issues

Test Engineer Co-op

January 2016 - May 2016

Space Exploration Technologies, Hawthorne CA

- Designed and built hardware to test bias-T HD cameras
- Created software to automate the creation of trouble reports
- Troubleshoot issues with avionics test racks

EDUCATION

Bachelor of Science, Computer Engineering

Rochester Institute of Technology, Rochester NY, Graduated May 2017

Digital Signal Processing

Cyborg Theory

Data Communication and Networks

HW & SW Design for Crypto Applications

PROJECTS

FPGA SDR RF Exploration (in progress)

- Building FPGA SDR exploration system to learn RF, VHDL, DSP, FPGA networking
- Initially using RF "lego" modules, transitioning to custom PCBs
- Created custom VHDL drivers for frequency generation, ADC, I2S
- Currently writing custom DMA SG drivers to send data to "real" computer for display

Home Networking (Homelab)

- Focus on internal and external services running on VMs, Linux, FreeBSD
- DIY'ed comms wiring, on and off-site backup, storage
- VLAN separation, hardware PCI passthrough, PoE

Yaesu VX-8 Battery

- Designed, built and tested a battery for a handheld Amateur Radio
- Designed a PCB that features USB-C charging, balancing and temperature monitoring
- Designed a 3D printed case featuring compliant clips, light pipes and a tight internal layout

Other Projects:

Soviet 120mm² RGBW VFD driver
Motorcycle Speedometer

Racecar Rollcage and Endurance Prep
Race Car Data logger and Comms Board

SKILLS & AWARDS

Eagle Scout
First Aid/AED/CPR, WFA, Stop the Bleed

Networking, subnetting, servers

Languages & Software:

C
Kicad PCB Design Software
Python
Linux/UNIX (Arch, Debian, BSDs)
VHDL

OpenSCAD and FreeCAD CAD Software
Wireshark
Microsoft Office
AutoIt Automation Scripting
Networking Equipment Configuration

PERSONAL INTERESTS AND HOBBIES

Amateur Radio Extra
Camping, hiking, and dancing
Community building
3D Printering

Photography (Analog, digital)
Holography (early stages)
Carbon fiber layup
Licked thing that was in space